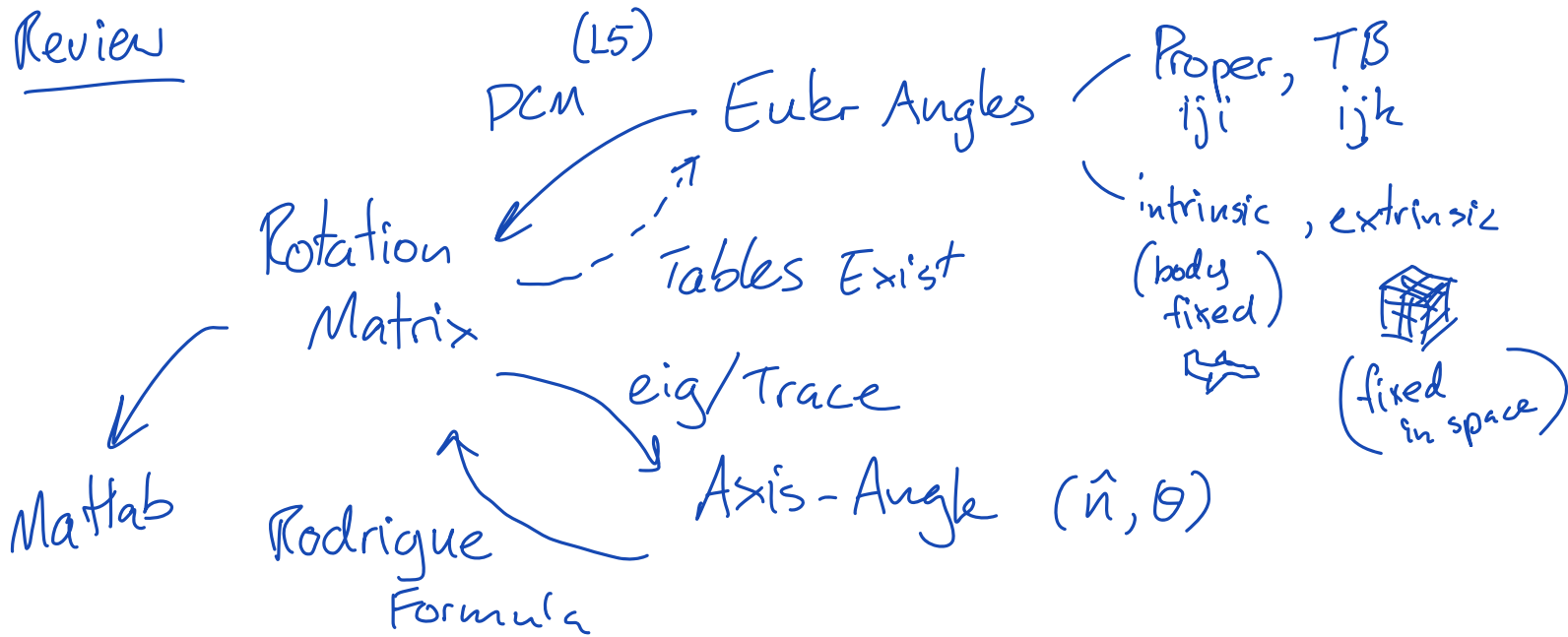


Axis-Angle Rotations

- Review
- Converting $RM \rightarrow AA$
- Converting $AA \rightarrow RM$
- Animation $\rightarrow$ see Canvas

HW2 Due Today
6pm.

Review



Converting Rotation Matrices to Axis-Angle

Euler Rotation Theorem

If R is a 3×3 matrix that is

(a) orthogonal $(RR^T = R^T R = I) \quad (R^T = R^{-1})$

(b) proper $(\det(R) = 1)$

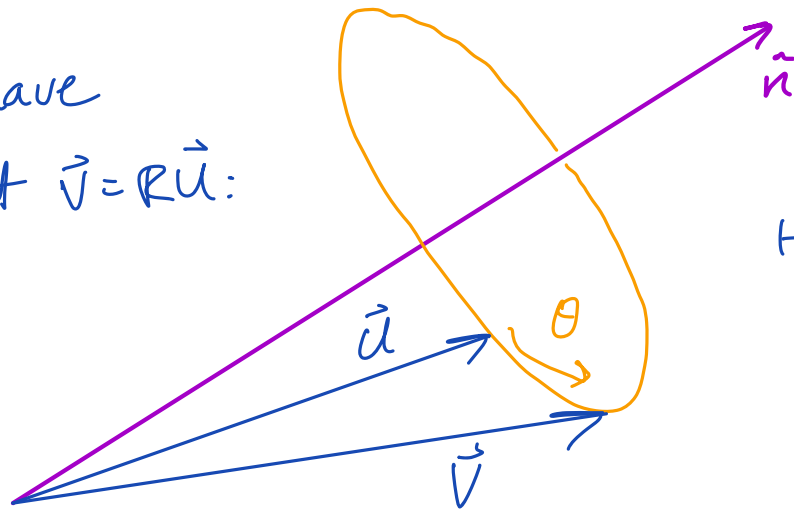
There must be a vector $\hat{n}$ such that $R\hat{n} = \hat{n}$

i.e. the rotation axis $\hat{n}$ is an eigenvector of R

[Review: 3Blue1Brown C14, Essence of Linear Algebra]

Assume we have

R such that $\vec{v} = R\vec{u}$:



How do we find $\hat{n}$ + θ ?

Matlab to the rescue

$$[A, b] = \text{eig}(R)$$

$\nwarrow$ eigenvectors $\swarrow$ eigenvalues

$$b = \begin{bmatrix} e^{i\phi} & 0 & 0 \\ 0 & e^{-i\phi} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

identify column with 1
 match to A

$$A = \begin{bmatrix} \{ & \} & \hat{n} \\ & & \hat{n} \end{bmatrix}$$

Notes: 1. $\hat{n}$ could be negative

2. Floating point $\neq$ integer $\Rightarrow \text{floor}(x)$

The angle θ can be found from the trace of R .

trace = Σ

$$R = \begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}$$

$$\text{trace}(R) = 1 + 2\cos\theta$$

$$\theta = \cos^{-1}\left(\frac{\text{trace}(R) - 1}{2}\right)$$

Note 3: $\cos(x) \rightarrow$ radians
 $\cosd(x) \rightarrow$ degrees

4. Will always be positive
 Be careful!

Converting Axis-Angle to Rotation Matrix

Use Rodrigues Equation

$$R = I \cos(\theta) + (1 - \cos(\theta)) \hat{n} \otimes \hat{n} + \mathcal{S}(\hat{n}) \sin(\theta)$$

where $\mathcal{S}(\hat{n})$ is skew symmetric matrix.

$$\mathcal{S}(\hat{n}) = \begin{bmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{bmatrix}$$

Example $R_x(90^\circ) \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$ $\hat{n} = [1 \ 0 \ 0]$, $\theta = 90^\circ$

$$R = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} \cos(90^\circ) + \underbrace{(1 - \cos(90^\circ))}_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} [1 \ 0 \ 0] + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \sin(90^\circ)$$

$$= 0 + 1 \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$





